

An Analysis of a NACA 0020 Aerofoil

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Abstract

The aerofoil lab, boundary layer lab, and finite wing lab allowed data to be obtained for the pressure distribution along the aerofoil as it changes with angle of attack, determine the viscous drag of the aerofoil and measuring the actual lift, and drag force of a half-wingspan. These results were then processed and compared to the theoretical values obtained in the CFD lab and the Xfoil and XFLR5 software, in order to determine the accuracy of the experimental process and theoretical analysis. Any discrepancies or mistakes are clearly analysed in this report, with results being discussed in the post process and clear conclusions are drawn on the overall accuracy and precision of the experiments.

Introduction

The NACA 0020 Aerofoil is an aerofoil with no camber and a maximum thickness at 20% of the chord length. The purpose of this rep is to determine the aerodynamic performance of the wing through both its lift and drag produced and measured, which are obtained through an aerofoil lab, a boundary layer lab, and a finite wing analysis lab. These will then be compared to the theoretical results and conclusions will be made on the accuracy of these results.

The key to processing the data obtained from the labs stems from the understanding of the theory behind it and implementing it correctly in situations where it pertinent to be used. For the aerofoil lab, the key underlying theories stem from thin aerofoil theory, such as the equation for obtaining C_l which is $C_L = 2\pi(\alpha - \alpha_0)$ [1]. Additionally, understanding of the theory behind modelling the aerofoil for Xfoil and CFD is needed in order to obtain accurate and relevant theoretical analysis.

For the boundary layer lab, an understanding of the transition from laminar to turbulent flow along the boundary edge as flow travels towards the trailing edge in conjunction with the Reynolds number for these distances from the leading edge is needed. The knowledge of Bernoulli's equation and calculating dynamic pressure is required to obtain the free stream velocity. This then allows for the local boundary layer thickness and displacement thickness to be calculated [2].

The finite wing lab builds on the knowledge required for the two previously mentioned labs, with the additional understanding that it may be necessary to model the wing as a rectangular wing for the equations given in the Jupyter notebook to apply.

XFLR5 requires an understanding of the impact that wing tip deflection causes and the disturbances in the wake and flow over the wing that this causes when comparing the theoretical data to the measured data.

Experimental and Numerical Methods:

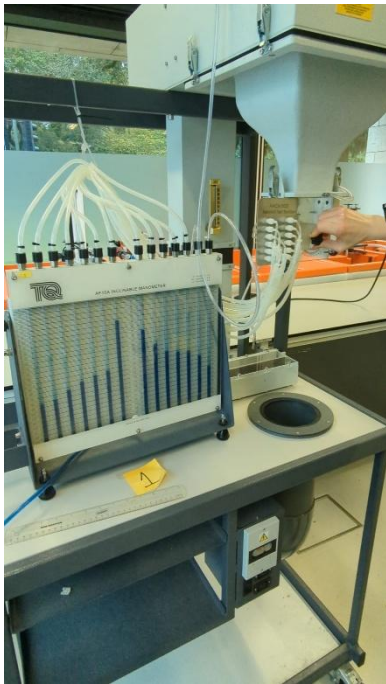


Figure 1: Boundary Layer Lab Setup

The boundary layer lab consisted of a plate with a 265mm long aerofoil, with height measurements of the manometer being taken at distances between 0.25mm to 10mm. These were done between Re of approximately 101000 and 490000, with distances from the plate being measured by a micrometer in 0.05mm increments and air temperature being measured by a thermometer. The pressure is measured by a fine pitot tube with a flat edge in order to push through the boundary layer [2]. To process the data, the calculated viscous drag profile of the aerofoil was plotted against the Re , allowing an estimate for total viscous drag to be obtained from the data on the graph. Uncertainty in the data obtained stems from the fact that both the pitot tube and the manometer were at a resolution of 1mm, so the closest resolution to the true value that could be measured was 0.5mm, which means that the overall absolute uncertainty for each measurement was ± 1 mm.

The aerofoil lab consisted of a NACA 0020 foil with a chord of 0.063m with 12 pressure tapings, 6 on the upper surface and 6 on the lower surface, which are connected to the same 12 columns on the manometer and the same wind tunnel as the boundary layer lab. The free stream had a velocity of approximately 20 ms^{-1} and measurements were taken with the angle of attack varying from -4° to 14° in 2° steps. The data was then plotted against the pressure distribution and integrated in order to obtain the lift curve slope which could be compared to the simulated data for further analysis and observation. Uncertainty here again stems from the fact that the resolution of the pitot tube and manometer are both 1mm so absolute uncertainty for each measurement is ± 1 mm. Additionally, the angles of attack were altered by hand to the resolution of 1° , so the percentage uncertainty is $\sim 8\%$, which could be a large cause for any discrepancies in the data during post processing.



Figure 2: Finite Wing Setup

The finite wing lab had a half-wingspan of 1.110m with a chord length of 0.400m, as seen in figure 2, with a flow speed of approximately 20 ms^{-1} , fluctuating between 19.96 ms^{-1} and 20.04 ms^{-1} . The lift and drag forces produced by the wing were measured using the weight balance in the wind tunnel at angles between -4° and 20° at intervals of 2° . From these data points, 8 are then selected to process and at lower angles of attack the flow is less turbulent and so is more likely to align with the simulated and theoretical data. Using the data, C_L and C_D are calculated for each angle of attack and plotted against each other, as well as the glide ratio vs C_L and C_L^2 vs C_D . These graphs were then compared to the CFD and XFOIL data to verify and observe any discrepancies. Uncertainty in the lift force measurement balance

is $\pm 0.10N$ which is a very small percentage uncertainty, as well as the low percentage uncertainty in the pressure measurements of the wind tunnel pitot tube so flow stream velocity had a very low percentage uncertainty. This meant that the data gathered for post processing from the wind tunnel was quite precise.

XFOIL was a programme that simulated the local pressure at 200 points on the NACA 0020 aerofoil, in order to generate a pressure distribution. The Aerofoil is then uploaded to the programme under similar conditions to the aerofoil lab, with a Re of 100000 and with viscous settings. The distributions are then obtained from angles of attack from -4° to 14° and then put into the Jupyter notebook to compare to the measured results. However, XFOIL notably tends to underpredict drag and overpredict lift unless flow is in a compressible range [3], so there are certainly discrepancies between gathered and simulated data during post processing as a result of this.

The CFD was done using Ansys which uses Reynolds-Averaged Navier Stokes equations in order to simulate the airflow over the aerofoil. The aerofoil geometry was uploaded such that the boundary conditions were not affected by the simulations. The analysis was carried out at a free stream velocity of 30ms^{-1} as was instructed, in order to ensure that there were no convergence issues. Although this increases the Re of the simulation, it negates the viscous effects on the simulation. Flow angle was altered rather than angle of attack in Ansys as this made the operation of carrying out the simulations far easier. Pressure distributions were calculated between 0° and 12° and then were processed in the Jupyter notebook to compare to the measured data obtained in the other labs.

XFLR5 worked with a 3D model of the wing as opposed to XFOIL, using XFOIL to find the pressure distributions but also taking into account the finite wing effect, allowing for data to be obtained that could be compared to the finite wing lab data. A chord length of 0.400m and wingspan of 2.200m was input at a free stream velocity of 20ms^{-1} to obtain pressure distributions at identical angles to the XFOIL data, which was then used to plot the same graphs as in the post processing of the XFOIL data. This means that the same XFOIL issue is present with the XFLR5 simulations, as well as the fact that XFLR5 models an entire wing as opposed to the measured data from a half wing model, so there is double wing tip deflection in the theoretical model, which is a potential issue in the post processing and so could be the reason for any discrepancy between both the XFOIL and XFLR5 data, and the simulated and measured data.

Results and Discussion:

Calculating Lift:

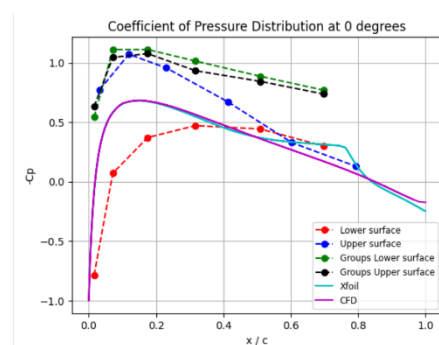


Figure 3: $-C_p$ vs x/c for $\alpha=0^\circ$

The aerofoil is symmetrical due to the fact that it has no camber, so it is expected that the upper and lower surfaces have identical pressure distributions and so no lift is produced at $\alpha=0^\circ$. This is because $\alpha=\alpha_0$ at this angle so the C_L theoretically should be 0. However, there is a clear difference in C_p along the position relative to the chord (x/c), with the values for lower and upper surfaces only truly converging at $0.5 x/c$. This misalignment with the simulated data may be due to

the fact that flow is assumed to be inviscid and irrotational, but real conditions are unable to replicate such ideal conditions.

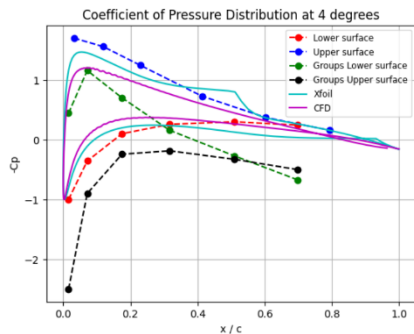


Figure 4: $-C_p$ vs x/c at $\alpha=4^\circ$

The data for the pressure distribution at an angle of attack of 4° shows how the C_p for the lower surface is higher than the C_p for the upper surface, hence lift is generated. As mentioned in the theory for the aerofoil lab, $C_L = 2\pi(\alpha - \alpha_0)$, where α is in radians. Using this theory against the value of C_L obtained from integrating between the curves allows for the comparison of experimental vs simulated vs theoretical data to be made. The theoretical C_L value is 0.439, while the experimental value obtained is 0.850 and the simulated value is 0.810. This gives an error of approximately 48.4% with the theoretical value, which is significant and is likely to be due to the fact that

flow was not inviscid. The experiment has an error of 4.8% compared to the simulated data, hence the results themselves are likely to be highly accurate. The experimental data gathered does agree with the theory and both upper and lower pressures converge at x/c of 0.6 where the simulations do as well. Therefore, the results are valid but ideal conditions are impossible to replicate and so the experimental C_L is likely to be almost half what ideal conditions suggest.

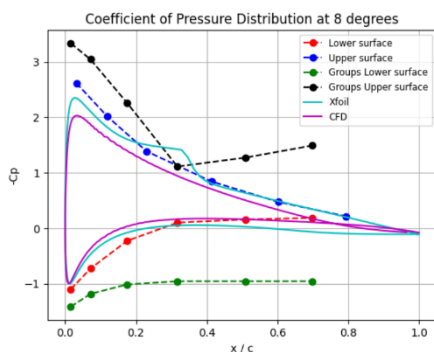


Figure 5: C_p vs x/c at $\alpha=8^\circ$

As was the case for an angle of attack of 4° , a similar trend for pressure distribution along the upper and lower surfaces of the aerofoil is expected with a larger difference between them, and this can clearly be observed due to the pressures converging further away from the leading edge of the aerofoil with a larger gulf in C_p compared to $\alpha=4^\circ$. The theoretical C_L was calculated to be 0.877, while the simulated C_L was 2.796 and the experimental C_L was 2.764. This means that again the experiment has clearly failed to hold up to the idealistic conditions required for the theory to completely hold up, as the percentage error is 68.3% which is quite significant. This is most likely due to the fact that flow is

definitely not irrotational and so tangential flows do not cancel each other out, leading to a much higher C_L being obtained than was expected. However, when compared to the simulated data, the percentage error is -1.16%, so the experiment itself is aligned with the expected results for true conditions and not idealistic ones.

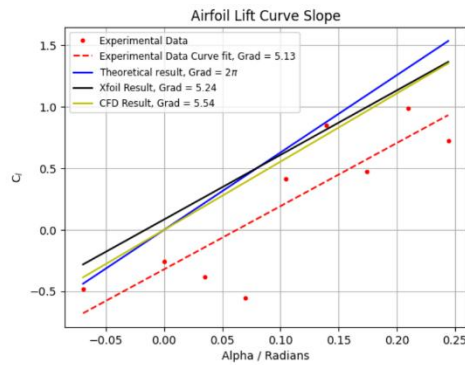


Figure 6 : Aerofoil Lab Lift Slope

As shown by the blue line in figure 6, the theoretical lift slope as predicted by thin airfoil theory is 2π and all 3 slopes fall below the theoretical gradient. However, they are all within 16% of each other and the predicted value, so results are valid. The only potential issue is the zero lift angle of attack for the data being -0.3 as can be seen from the data in figure 6, which can be due to a zero error in reading of any of the measurements or a systematic error in the wind tunnel or the manometer.

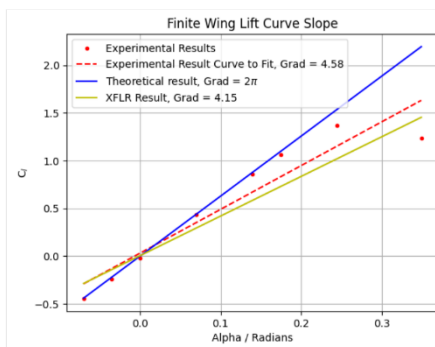


Figure 7: Finite Wing Lab Lift Curve Slope

The finite wing produced a lift curve slope that was further away from the theoretical value of 2π and had a percentage error of 9.39% from the simulated lift curve slope so the analysis of the wing holds quite well between the experimental and simulated data – the theory fails to hold up with every value of C_L calculated due to the fact that flow cannot be modelled as idealistic in the real world and so discrepancies are naturally going to be present.

Calculating Drag:

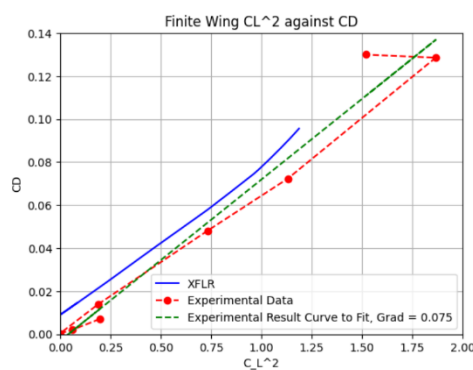


Figure 8: Cd vs Cl^2

Figure 8 allows zero lift drag to be obtained from the y intercept of the graph and allows k – lift induced drag - to be obtained from the gradient as $C_d = C_{d0} + k C_l^2$ [2]. As seen in the graph, $k = 0.075$ in the experimental data whereas for XFLR5 it is -0.175 , which is a percentage error of 333%, so clearly the data has had clear random errors as can be seen from the graph. XFLR5 simulates a C_{d0} of 0.012 as opposed to the obtained value of -0.004 from the experimental data, which is a percentage error of 400%. Therefore, it is clear to see that the finite wing lab is not reliable in drawing conclusions for the drag produced by the wing, likely to be due to the flow not being of a similar velocity to the simulated data.

Figure 8 also displays the C_f of the aerofoil for the laminar and turbulent boundary layers which can be obtained from the graph. The data set is very similar to the theoretical C_f for a turbulent boundary layer, so it is very likely that the boundary layer of the wing was turbulent.

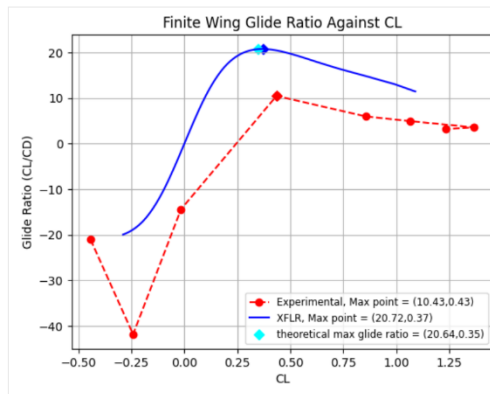


Figure 9: Finite wing glide ratio

Theory predicts an inverted parabolic shape for the relationship of the glide ratio versus the C_L , peaking at a C_L of approximately 0.35. The experimental data had one clear anomaly at a C_L of -0.25 but overall followed the trend suggested by XFLR5, with a percentage error of 18.6% in the C_L and 98.7% in the actual glide ratio, so while the data follows the theory and simulations, the values obtained are almost half the predicted values, possibly due to the fact that the finite wing lab only used a half wing model.

Conclusions:

In terms of calculating lift, experimental data was quite accurate as it was generally within 10% of the simulated predictions and so the experiments were useful in obtaining relevant and accurate data. The curves followed the predicted trends and the slopes were close so it can be concluded that the simulations and experiments worked well in conjunction to accurately determine the coefficients of lift for different angles of attack. The theory held generally in terms of expected behaviour but was almost always off by at least 100% of the experimental value, so it was probably due to a use of a half wing model in the finite wing lab.

For calculating drag, the data seemed to be more inaccurate especially due to the fact that values were so small so any fluctuations in the data recorded due to a random error were likely to influence the data a lot. This led to very inaccurate values for C_f and C_{d0} being obtained from the graphs. These experiments would need to be repeated in order to actually determine whether the experiments were a valid method to determine the drag of a NACA 0020 aerofoil or whether it was simply down to a freak random error in the wind tunnel.

Overall, the experiments followed the expected patterns and behaviours and somewhat accurately predicted the coefficients for lift and drag so the aerofoil is relative efficient in its glide slope, lift produced and drag.

References

- [1] B.Ganapathisubramani, S.Krishna, (2024), Aerofoil Lab Jupyter Notebook
- [2] B. Ganapathisubramani, S.Krishna, (2024), Boundary Lab Jupyter Notebook
- [3] Kristen Kallstrom, (2022), Exploring Airfoil Table Generation using XFOIL and OVERFLOW [Online] Available at:
https://rotorcraft.arc.nasa.gov/Publications/files/Kristin_Kallstrom_Final_Paper_18-Jan-2022.pdf